LIDZHADE RUDZAMI

Research Assignment 1

1. A database is a digital respository for storing, managing and securing organised collections of data.

Eg 1. E-commerce : When we search for a product on platforms like takealot or Amazon, a database instantly pulls up live inventory, price and customer reviews to display on our screens

2. Banking : Every time when we swipe our cards or transfer money, banking database securely records our account balance in a fractions of a second to prevent fraud and ensure accuracy

Q2. A database stores data securely, while a data-warehouse stores data for analytical processing.

Why companies use a data warehouse?

→ They use data warehouse to centralise scattered information into a single ~~respo~~ repository, allowing them to analyze historical performance and make data driven decisions

Q3. A Data lake is a centrallised storage respository that holds vast amount of raw, unprocessed data in its native format

Data Lake – Data is stored in its raw form. Stores anything

Data warehouse – Data must be cleaned. Stores highly curated, relational data.

→ ~~when~~ A data lake is an ideal choice for a business with a massive volume of data and it does not require upfront data cleaning or restructuring.

Q4. A data mart is a subset of a data warehouse focused on a particular line of business, department or subject area. It can be used by a specific department, team or business use case.

Q5. Structured data is highly organised and easy to search unstructured data is free-form and complex
Semi-structured acts as a flexible middle ground.

Structured: organises info into rows and columns using a fixed schema (Bank transactions)
Unstructured: No predefined schema or format (JSON files)
Semi-structured: Some structure with flexibility, using tags or key-value pairs to organise data. (video files)

Q6. Data modeling is the process of creating a visual blueprint of an information system defining how data is collected, structured and connected.

Why is data modeling important?

- Ensures Data intergrity
- Saves time and money
- Improves communication

Types of Data Models

- Logical Data model
- Conceptual Data model

Q7. Data mining is a process used by companis to turn raw data into useful information.

How is it used:

- customer segmentation and personalization
- Fraud Detection
- Churn prediction

Two examples

- Netflix or youtube can use classification algorithms to understand viewers habits.
- Retails like Amazon uses prediction mining techniqus to generate personallzed shopping recommendations.

Q8. DBms, is similar to a central, super-organized File cabinet where employees can securely access and manage company data and the databases that stores that data without having to worry about where the data is located or what format it is in.

- Cloud-based DBms (Snowflake and Data bricks) Single place for organizations to store, use and share their data.
- Relational DBms (mySQL), stores data in tables of row and columns that are fairly rigid.
- No SQL DMBms (mongo DB) - Are best suited for data structures that have loose or changing definitons and conte

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Q9. ETL → is a core data intergration process used to gather raw data from multiple disjointed sources, clean and structure it. It gets loaded into a centralized system like a data warehose or data lake

1 ~~Extract → extracted raw data~~

1 Extract → Data is retrieved or copied from disparate source systems and temporarily stored in a staging

2 Transform → Extracted raw data gets processing, cleaning and filtering.

3 Load → Cleaned and transformed data is finally stored into a target system like a centralised cloud database.

Q10. Common Data Formats

CSV Format. is a text based tabular data format. Each line represents a row of data and fields in a row a separated by commas.

JSON (Javascript object notation)

It is a lightweight text based data format for storing and sharing data, humans and computers can read it easily

XML File-

It is great for display data and it is also a text based data format